Problem Chosen	2024 ShuWei Cup Summary Sheet	Team Control Number
D		2024092929287

summary

This article focuses on two cities, based on data mining of housing information and service levels in the appendix, and has established mathematical models for prediction, estimation, and comprehensive evaluation, providing housing price forecasts, housing stock data, and conducting urban advantage and disadvantage analysis as well as resilience assessment.

For Question 1, after analyzing the data in the appendix, we decided to establish four predictive models (linear regression, decision tree, random forest, and gradient boosting). We preprocessed the data for missing values, trained the above four models, and validated them with the R2 metric, concluding that the gradient boosting regression model performed the best, with an R2 coefficient of 0.9269, which is closer to 1 and indicates the strongest correlation. The gradient boosting regression model was then applied to regional housing price prediction. The results were visualized, and the total housing stock was estimated.

For Question 2, this article conducts a quantitative analysis of the service levels across fifteen different industries in City 1 (Changchun) and City 2 (Hohhot), establishing a multi-indicator comprehensive evaluation method, resulting in a score of 4725.87 for City 1 and 2801.21 for City 2, suggesting that City 1 has a higher overall service level. By visualizing the comparison of the number of service industries in the two cities, we identify commonalities and unique characteristics between the two cities and analyze their respective strengths and weaknesses.

For Question 3, in order to invest in the two cities, we established a multi-indicator comprehensive evaluation model, using the provided fifteen documents (service indicators), preliminarily determining the correlation within each category of data, identifying indicators that contribute to measuring the resilience and sustainability of the cities, and quantitatively assessing the sustainable development capabilities of the cities.

For Question 4, based on the conclusions drawn from Questions 1 to 3, we summarized the strengths and weaknesses of the two cities and listed key investment areas, investment amounts, and expected improvements in smart city development levels. This aims to further develop strengths and continuously address weaknesses.

Key word: Big Data Statistical Analysis,Random Forest,Decision Tree,Resilience Evaluation,MCE

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1. Introduction

1.1 Background

In recent years, with the intensification of China's population aging trend, global climate change, and challenges such as economic stagnation, especially since the pandemic, urban resilience and sustainable development capabilities have increasingly become key indicators of urban construction. Despite these challenges, China's urbanization process continues to advance, with growing demand for urban housing. Meanwhile, housing price fluctuations have become critical factors affecting residents' quality of life and social stability. At the same time, the government is raising requirements for improving urban infrastructure and service levels to address complex social and economic environments.

The continuous growth in housing demand prompts a reevaluation of current urban planning and housing supply structures. Balancing the relative stability of housing prices while meeting citizens' housing needs has become a focus for policymakers and researchers. In recent years, as urban populations grow and economic development drives diverse demands, the improvement and enhancement of urban service systems have gradually been placed on the agenda. Quantifying and analyzing service levels across cities and assessing their emergency response capabilities to extreme situations such as natural disasters and climate change have become pressing issues in urban management.

Against this backdrop, it is necessary to build scientific models to predict future urban housing price trends, estimate housing stock, and conduct comprehensive analyses of service levels and resilience. This approach will help formulate actionable policy recommendations, improve cities' risk resistance, and promote sustainable economic development.

1.2 Problem Review

Question 1: Develop an accurate and effective predictive analysis model to forecast the trends in housing prices for City 1 and City 2 over the next few years. Incorporate economic factors such as population and GDP to estimate the scale of the existing housing stock. Based on the analytical and predictive results, propose relevant urban housing policy recommendations to meet residents' housing needs.

Question 2: To analyze the current state of urban service systems, perform a quantitative analysis of multiple service domains such as healthcare, education, and living facilities using various service indicators. Identify the strengths and weaknesses of the service systems in both cities and propose improvement suggestions to enhance service levels.

Question 3: To enhance urban resilience to extreme events and sustainable development capabilities, propose a resilience evaluation method based on multiple indicators. Assess the emergency response capabilities and sustainable resource utilization in both cities to identify critical areas for development.

Question 4: Combine the findings from the above questions to provide recommendations to urban management departments for optimizing resource allocation and development planning.

2. Problem analysis

2.1 Analysis of question 1

For Question 1, the first step is to conduct proper data preprocessing. Then, three models (Multiple Linear Regression, Random Forest, and Gradient Boosting) are used for predictive modeling. After training the models, the one with the best performance is selected as the final model for regional housing price prediction. Finally, visualization is conducted, and the total housing stock is estimated.

2.2 Analysis of question 2

For Question 2, the first step involves proper data preprocessing, followed by using a Multi-Criteria Evaluation (MCE) method for comprehensive evaluation. This approach provides service level scores for each city across various domains, enabling a ranking and comparison of overall service levels. Finally, visualizations can be created to display the geographic distribution of different service types (latitude and longitude) for the two cities.

2.3 Analysis of question 3

The analysis of Question 3 focuses on comprehensively assessing the emergency resilience and sustainable development levels of city1 and Hohhot. This evaluation is centered on key areas, including the distribution and accessibility of transportation, medical, and public facilities, which are crucial for emergency response. It also involves retail, lifestyle, and government services to gain insight into the cities' basic supply guarantees and coordination capabilities during crises.
The model identifies weaknesses in the urban systems and proposes short-term and long-term investment strategies, with an emphasis on strengthening infrastructure in vulnerable areas, optimizing transportation networks, and developing sustainable systems. These measures aim to enhance overall resilience and ensure balanced development of the cities under financial constraints.

2.4 Analysis of question 4

Urban development requires addressing three key challenges: fluctuations in housing demand, optimization of service systems, and resilience building. Through predictive models and indicator evaluations, the weaknesses of both cities can be quantified, enabling targeted optimization strategies. Combining short-term improvements with long-term planning will enhance urban competitiveness and sustainable development capacity.

3. Model Symbol and Assumptions

3.1 Symbol Description

To facilitate the model construction and solving process, the key symbols used are explained as follows:

Symbol	Meaning
A	Area of Coverage
N	Number of Service Points
Density_Index	Density Index
$\max(\text{lon}), \min(\text{lon})$	Maximum and Minimum Longitude
$\max(lat), \min(lon)$	Maximum and Minimum Latitude
S_i	Comprehensive service level score for city i
W_k	The weight of the k-th indicator
x_{ik}	The score of city iii for the k-th indicator
n	The total number of indicators
μ, σ	Mane and Standard Deviation
d_{ij}	The difference in ranks between the feature and the target variable for the sample
X'	the feature value of the sample
Y	Target vector for housing prices
X	Feature matrix
β	Regression coefficient vector, including the intercept
ε	Error term vector
B	the number of decision trees
$\bar{y}$	the actual target values

3.2 Problem assumptions.

Reliable Data: It is assumed that the data is accurate and reliable, reflecting key factors necessary for building the models.

Service Level Quantification: Service levels are quantified based on facility conditions. Urban services are influenced by the distribution and number of facilities, and the demand structures of residents in the two cities are assumed to be similar.

Short-Term Stability of Facility Impact:It is assumed that the influence of facilities on housing prices and the return on investment in various sectors remains stable over the short term.

Quantifiable Resilience to Extreme Events: The resilience of facilities to extreme events is assumed to be measurable, with the nature of extreme situations in both cities considered similar and predictable.

Limited Fiscal Resources and Resource Optimization: It is assumed that city fiscal resources are limited, with varying returns on investment across sectors. Optimal resource allocation is necessary, and resilience is positively correlated with sustainable development.

Perfect Sphere: Assuming the Earth is a perfect sphere, for small areas (such as a city), the area is considered to be approximated as a rectangle.

4. Model Solving and Analysis

4.1 Question 1 Model Analysis and Solving

The objective of Question 1 is to provide specific predictions for future housing prices in City 1 and City 2, as well as to estimate the existing total housing stock. To achieve this, we employed multiple machine learning regression models for housing price prediction. The process includes the following steps: data preprocessing, feature selection, model construction and training, model prediction, and result evaluation and analysis.

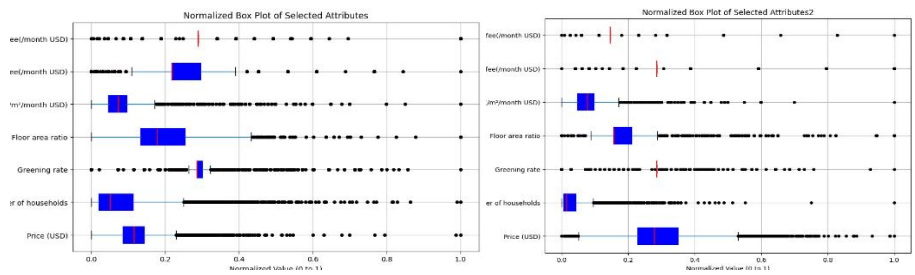
4.1.1 Data Preprocessing

Based on the basic information of properties for sale in City 1 and City 2 retrieved from 58.com (Appendix 1 and Appendix 2), a significant number of missing values (N/A) were identified. The missing values for the feature variables are visualized in picture 1.

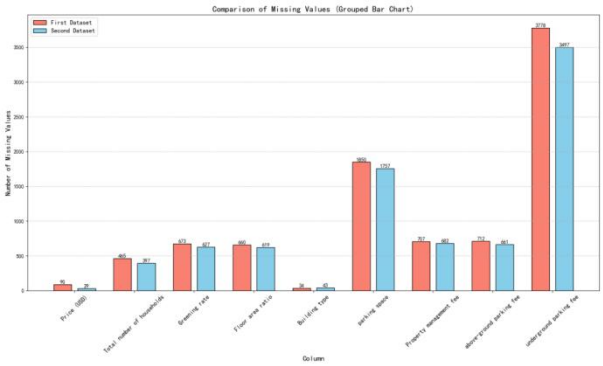
KNN Interpolation: Missing values were replaced and filled using KNN (K-Nearest Neighbors) interpolation.

Numerical Variables: Missing values in numerical variables were filled using predictions generated by Random Forest.

Categorical Variables: For categorical variables, missing values were replaced using the mode of the column.



picture 1,picture 2 City1,2 Outlier Detection



Picture 3 Comparison of Missing Values Between the Two Cities

Data Encoding and Standardization: Label Encoding: For categorical variables (e.g., Building type, Property type), one-hot encoding (0/1) is used to convert them into numerical representations.

Min-Max Normalization: Standardizes features and target variables by scaling all values to the range of [0, 1]. The formula is as follows:

$$x_{ij}^{\text{标准化}} = \frac{x_{ij} - \mu_j}{\sigma_j}, \quad \forall i \in \{1, 2, ..., n\}, \quad \forall j \in \{1, 2, ..., p\}$$

• x_{ij} represents the j feature value of the i sample. The mean μ_j and standard deviation σ_j are defined as follows:

$$\mu_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}, \quad \sigma_j = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \mu_j)^2}$$

4.1.2 Model Building

By evaluating the correlation between each feature and the target variable, features that are highly correlated with house prices are selected for modeling. For both the features and the target variable, their values are ranked to obtain the ranks. Ranking involves sorting all the values of a feature in ascending order, and similarly for the target variable.

Definition of Spearman's Rank Correlation Coefficient:

$$\rho_j = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}, \quad d_i = R(x_i) - R(y_i)$$

d_i : The difference in ranks between the feature and the target variable for the i sample.

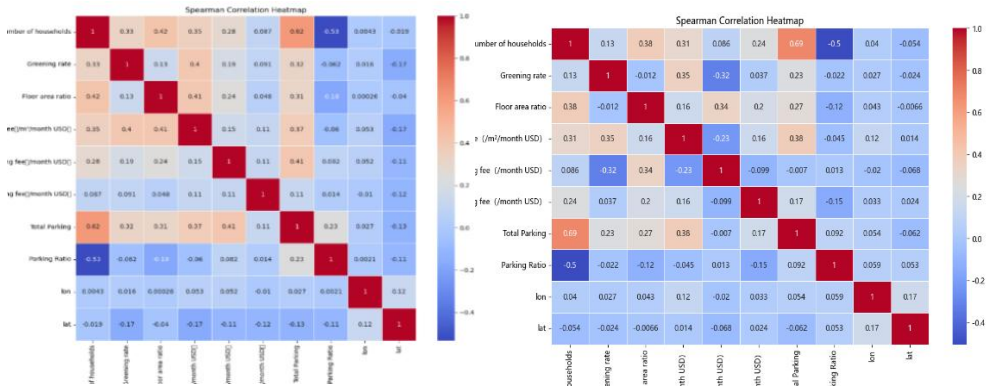
n : The total number of samples.

Feature Selection:Select the features corresponding to the largest absolute values of

ρ_j as the final input features for modeling.

4.1.3 Model Solving

Conduct Spearman correlation analysis on the feature variables in Appendix 1.xlsx and Appendix 2.xlsx using Python, as follows:



Picture 4 city1 and 2's Correlation Results

The correlation between total users and total parking spaces is 0.62, with a correlation of 0.53 with the parking space ratio and 0.42 with the floor area ratio (FAR), indicating a strong positive correlation.

It has a certain degree of positive correlation with "Property management fee (/m²/month USD)" and "Above-ground parking fee (/month USD)", which might suggest that areas with higher FAR have common characteristics in terms of property management and parking fees.

The greening rate shows relatively weak correlations with other variables:

A slight positive correlation with "Floor area ratio" (0.13).

A weak positive correlation with "Total Parking" (0.32).

No significant correlations with other variables.

This indicates that the greening rate is relatively independent of other key features in the dataset.

4.1.4 Model Comparison

(1)Multiple Linear Regression Model Construction

Based on the feature selection from Spearman correlation, the linear relationship between housing prices and the selected features is fitted. The optimal coefficients are solved using the least squares method.

The model is constructed as follows:

$$\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{\epsilon}$$

$\mathbf{y} \in R^n$:Target vector for housing prices.

$\boldsymbol{x} \in \boldsymbol{R}^{n\times 10}$:Feature matrix.

$\boldsymbol{\beta} \in \boldsymbol{R}^{10+1}$:Regression coefficient vector, including the intercept.

$\boldsymbol{\epsilon} \in \boldsymbol{R}^{n}$:Error term vector.

Minimizing the sum of squared residuals, the loss function is defined as:

$$\boldsymbol{J}(\boldsymbol{\beta}) = \sum_{i=1}^{n} (\boldsymbol{y}_i - \hat{\boldsymbol{y}}_i)^2 = \sum_{i=1}^{n} (\boldsymbol{y}_i - \boldsymbol{x}_i^T \boldsymbol{\beta})^2$$

Based on the least squares method, the loss function is established and optimized, leading to the closed-form solution:

$$\boldsymbol{\beta} = (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{y}$$

(2)Random Forest Regression Model Construction

Random Forest combines multiple decision trees into an ensemble model to capture the nonlinear relationships between features and the target variable.

$$\hat{y}_i^{(t)} = f_t(x_i), \quad t \in \{1, 2, \ldots, T\}$$

where T is the number of decision trees.

The prediction of the Random Forest is the average of the predictions from all trees:

$$\hat{y}_i = \frac{1}{T} \sum_{t=1}^{T} \hat{y}_i^{(t)}, \quad i \in \{1, 2, \ldots, n\}$$

By introducing an ensemble of multiple trees, Random Forest can effectively capture complex nonlinear feature relationships and reduce model variance.

(3)Gradient Boosting Regression Model Construction

By iteratively fitting residuals, Gradient Boosting constructs decision tree models step by step, improving predictive performance.

• Step 1: Initialize the Model

$$F_0(x) = \arg\min_{\gamma} \sum_{i=1}^{n} L(y_i, \gamma)$$

Typically, the mean value of the target variable is chosen as the initial model.

• Step 2: Selection Process：In each iteration (or round), the goal of the mmm-th iteration is to fit the residuals, defined as:

$$r_i^{(m)} = -\frac{\partial L(y_i, F_{m-1}(x_i))}{\partial F_{m-1}(x_i)}$$

Construct a new decision tree $h_m(x)$ to fit the residuals $r_i^{(m)}$。

• Step 3: Model Update

$$F_m(x) = F_{m-1}(x) + v \cdot h_m(x)$$

Here, $v \in (0, 1]$ is the learning rate, controlling the contribution of each tree to the model and helping to improve the model's convergence efficiency.

Once the model is built, predictions are made for each region. For each region (determined by citycode and adcode）evaluations are conducted to obtain prediction results.

（4）Model Evaluation and Validation

Model generalization ability is improved through K-fold cross-validation, and the best model is selected using evaluation metrics.

K-Fold Cross-Validation：The dataset is divided into 10 parts (10-fold cross - validation). For each round, 9 parts are used for training, and the remaining 1 part is used for testing. This process is repeated 10 times.

Cross-Validation MSE:

$$CV_{ERROR} = \frac{1}{10} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i=1}^{n_k} \left(y_i - \hat{y}_i \right)^2$$

Mean Squared Error (MSE):

$$\boldsymbol{MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}$$

Coefficient of Determination (R^2):

$$\boldsymbol{R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}_i\right)^2}}$$

$\bar{y}$ is the mean of the actual target values.

4.2 Question 2 Model Analysis and Solving

The goal of Problem 2 is to conduct a quantitative analysis of multiple service domains, such as healthcare, education, and living facilities, based on various service indicators. This involves identifying the strengths and weaknesses of the service systems in the two cities and proposing improvement suggestions to enhance service levels. To achieve this goal, the Multi-Criteria Evaluation (MCE) method is used for analysis and evaluation. The specific steps include data preprocessing, feature selection, model construction and training, model prediction, result evaluation, and analysis.

4.2.1 Data Preprocessing

Area Calculation: Assuming the Earth is a perfect sphere, for small regions (e.g., a city), the area can be approximated as a rectangle. The formula is as follows:

$$Area = (\max_lon - \min_lon) \times (\max_lat - \min_lat)$$

Density Index: Describes the density of a specific factor (e.g., population or facilities) in a region. It is calculated by dividing the number of specific factors by the area of the region:

$$Density_Index = \frac{N}{Area}$$

4.2.2 Model Building

(1) Coefficient of Variation Method

The coefficient of variation method is an objective weighting method that assigns weights to indicators based on their variation between current and target values. Greater variation indicates more difficulty in achieving the target, and thus the indicator should be assigned a higher weight.

Calculation steps:

Step1）Mean Calculation:

$$\mu = \frac{1}{\mathrm{n}}\sum_{i=1}^{n} x_i$$

Step2）Standard Deviation Calculation:

$$\sigma = \sqrt{\frac{1}{\mathrm{n}}\sum_{i=1}^{n}(x_i - \mu)^2}$$

Step3）Coefficient of Variation Calculation:

$$CV = \frac{\sigma}{\mu}$$

(2)Multi-Criteria Evaluation (MCE)

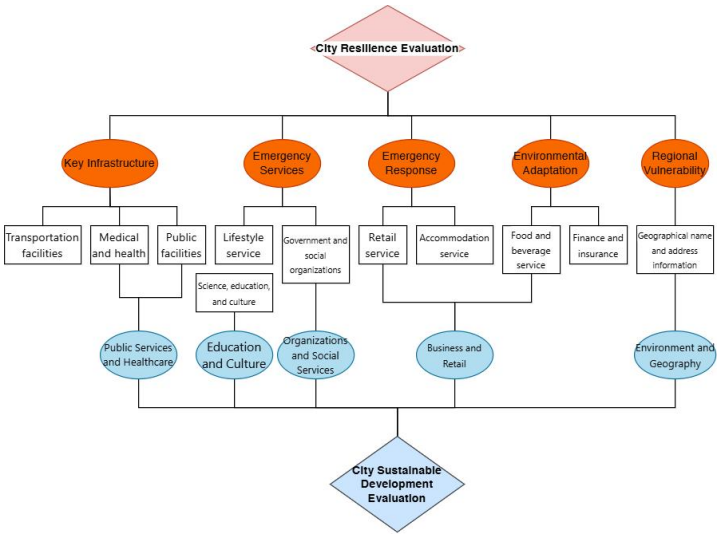
MCE is a widely used mathematical method for evaluating urban service levels. By aggregating multiple indicators with assigned weights, it calculates a comprehensive evaluation score to quantitatively analyze service levels. Indicators can cover various domains of public services, such as education, healthcare, transportation, and environmental quality.

For instance, to evaluate the performance of City 1 and City 2 in different domains, assume the indicators are$x_1, x_2, x_3 \ldots\ldots x_n$, The overall service level score for a city is calculated as:

$$S_i = \sum_{k=1}^{n} W_k \times x_{ik}$$

4.3 Question 3 Model Analysis and Solving

Based on the data from Appendixes 3 and 4, conduct a multi-faceted evaluation of the resilience of city1 and Hohhot in responding to extreme weather and emergencies, as well as their sustainable development capabilities. This evaluation will provide a foundation for urban planning and development.



picture 5 City Evaluation Framework Diagram

4.3.1 Key Infrastructure

(1)Transportation Facilities (Transportation facilities data.csv): Layout of the transportation network and distribution of key facilities. The rationality of hubs and emergency corridors to ensure emergency transport and evacuation.

(2)Medical and Health (Medical and health data.csv): Geographical distribution of medical resources, with assessments of accessibility and emergency capacity.

(3)Public Facilities (Public facilities data.csv): Distribution of various public facilities. Suitability and functional completeness of shelter and rescue facilities.

4.3.2 Emergency Services

(1)Lifestyle Services (Lifestyle service data.csv): Layout of lifestyle facilities and operational guarantees during extreme weather. Emergency response mechanisms and material supply capabilities.

(2)Government Organizations (Government and social organizations data.csv):

Allocation and distribution of departmental and organizational resources. Evaluation of emergency coordination and preparedness.

4.3.3 Emergency Response

(1)Retail Services (Retail service data.csv): Distribution of stores and stability of material supplies during extreme weather. Recovery speed after disasters and the ability to meet residents' needs.

(2)Accommodation Services (Accommodation service data.csv): Quantity and distribution of accommodation resources, post-disaster accommodation capacity, and suitability and functionality of shelters and temporary accommodations.

4.3.4 Environmental Adaptation

(1) Food and Beverage Services (Food and beverage service data.csv): Distribution of dining facilities and food supply security. Fulfillment of social responsibility and participation in emergency supplies.

(2)Finance and Insurance (Finance and insurance data.csv): Distribution of financial facilities and assessment of cash flow. The role of insurance services in risk-sharing and economic compensation.

4.3.5 Regional Vulnerability

(1)Geographical Name and Address Information (Geographical name and address information data.csv): Regional division analysis, incorporating natural, social, and economic factors. Identification of vulnerable areas and provision of disaster prevention and mitigation references, as well as recognizing regions prone to extreme weather impacts.

4.3.6 Urban Sustainable Development Evaluation Dimensions

Infrastructure and Transportation: Based on Transportation facilities data.csv, analyze transportation networks, including stations, hubs, and roads, to assess convenience and capacity. Integrating motorbike and car data, evaluate private transportation penetration and dependence to determine transportation sustainability.

Public Services and Healthcare: Using Public facilities data.csv and Medical and health data.csv, evaluate the accessibility of public services, healthcare coverage, and fairness in resource allocation, given their relevance to residents' quality of life and health, influencing urban sustainability.

Education and Culture: Leveraging Science, education, and culture data.csv, assess the diversity of educational and cultural facilities and their impact on improving residents' skills and knowledge, contributing to sustainable urban development.

Business and Retail: Utilizing Retail service data.csv and Food and beverage service data.csv, analyze commercial density, consumer purchasing power, and patterns to reflect urban economic vitality and sustainable consumption.

Organizations and Social Services: Based on Government and social organizations data.csv, evaluate government service coverage and the role of social organizations in promoting sustainable development.

Environment and Geography: Using Geographical name and address information data.csv, analyze urban geographical distribution and regional division to understand resource strengths and weaknesses, providing a foundation for sustainable development planning.

Table 1 Weight Distribution of City Resilience Evaluation Indicators

CityResilience Evaluation Indicators	Weight (%)	Reason
Transportation Facilities	20	As the 'artery' of urban emergency response, its layout impacts the efficiency of material transport and evacuation, providing critical support for emergency rescue.
Medical Facilities	20	Critical for safeguarding residents' health during and after emergencies, with high importance in post-disaster recovery phases.
Public Facilities	15	Includes shelters and emergency power supplies, essential for basic living security after disasters, though less central than transportation and medical facilities.
LifestyleService Facilities	10	Meets residents' basic living needs during emergencies, assisting in maintaining social order, but not directly involved in rescue efforts.
Government and Social Organizations	10	Responsible for coordination, resource allocation, and communication during emergencies, essential for managing crises effectively.
Food and Beverage Services	10	Provides essential sustenance for residents and rescue personnel,crucial during prolonged emergencies but secondary to infrastructure and coordination.
Financeand InsuranceFacilities	5	Plays a role in post-disaster economic recovery, with limited direct impact during emergencies.
Accommodation Services	10	Offers shelter and housing for affected individuals, important for stabilization and reconstruction efforts after disasters.

5.Test the model

5.1 Test the model of question 1

By writing Python code, we calculated the MSE and R^2, as shown below:

Table 2 Model Evaluation Results

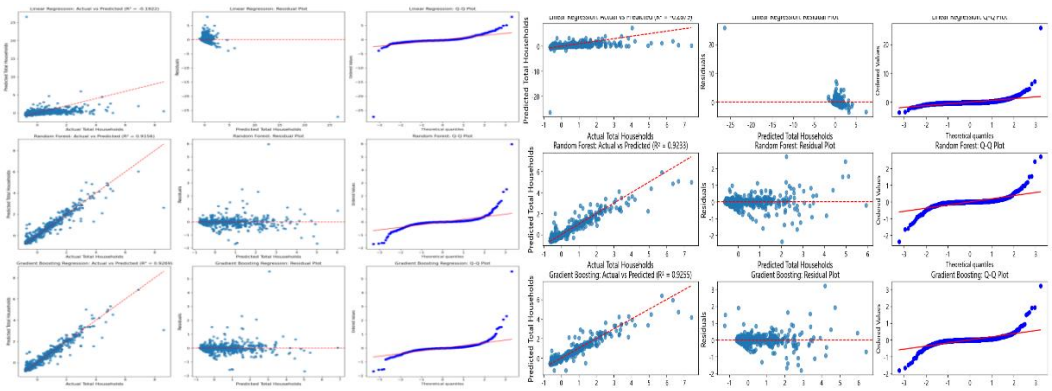
ModelEvaluation	MultipleLinear Regression	RandomForest Regression	GradientBoosting Regression
MSE	1.1465	0.0811	0.0703
R^2	0.1922	0.9156	0.9269
Cross-Validation R^2	0.1406	0.9258	0.9285

Multiple Linear Regression Formula：Based on Table 2, the Multiple Linear Regression model has $MSE = 1.1465$, $R^2 = -0.1922$,The regression formula derived using Python is as follows:

$$y_1 = 0.0058 + 0.1994x_1 + 0.0816x_2 + 0.0518x_3 + 0.5771x_4 - 0.0579x_5$$
$$y_2 = 0.0058 + 0.1994x_1 + 0.0816x_2 + 0.0518x_3 + 0.5771x_4 - 0.0579x_5$$

Where $x_i\,(i=1\ldots5)$: Floor area ratio、above-ground parking fee、underground parking fee、Total Parking、Parking Ratio

5.1.1 Model Visualization and Comparison



Picture 6 city1 and 2's model test

Linear Regression:poor performance.R2=0.1922(a negativeindicates the model performs worse than using the mean for prediction).Actual and predicted values show significant mismatch.Residuals are not effectively distributed, indicating the model captures few data characteristics.

Random Forest Regression: Good fit. R2=0.9156, with most scatter points near the ideal fit line.Residual Characteristics:Residuals are randomly distributed but show some bias at extreme values.

Gradient Boosting Regression (GBR): Best.R2=0.9269, outperforming the other models.Residual Characteristics: Residuals show good normality and randomness, closely following a normal distribution, making this model effective at capturing complex nonlinear relationships.

Conclusion: The Gradient Boosting Regression model demonstrates the best performance, making it suitable for predicting urban housing prices.

5.1.2 Estimating Total Housing Stock

In the absence of precise housing area data, the Total Housing Stock(THS) is often estimated using the total number of households and the average area per household.

Total Households(TH):This data can be obtained from census or housing statistics, which reflect the urban housing capacity.

Average Area per Household:When precise data is unavailable, **the Floor Area Ratio (FAR)** can be used as an approximate substitute. Although not entirely accurate, this approach is feasible when data is lacking.

This method provides a foundational approximation for relevant research in the absence of comprehensive data.

$$THS = TH \times FAR$$

5.1.3 Model Results

Through calculations, the total existing housing areas in the two cities are determined to be 5,307,572.93 m² and 2,878,345.76 m², respectively. Additionally, predicted future housing prices for various regions are provided (the complete dataset

is stored in the problem1.xlsx file), and the full results are visualized.

Table 3 Linear Regression Model Results

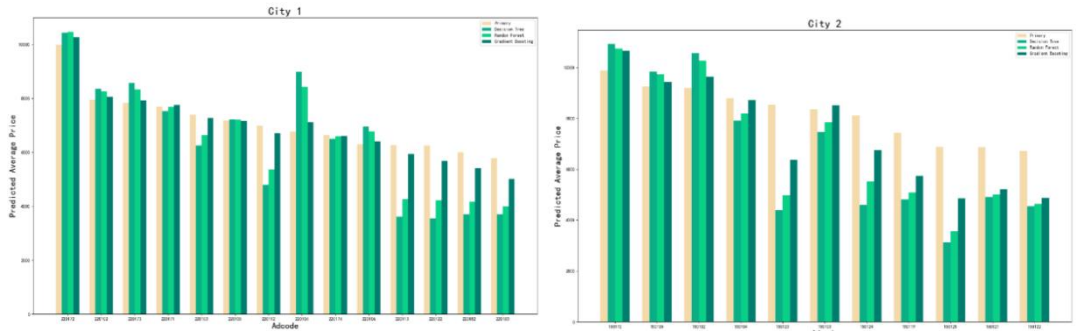
Citycode	Adcode	Predicted Average Price (USD)
431	220104	6764.782218
431	220173	7825.148383
431	220103	7395.310954
431	220106	6301.285135

Table 4 Random Forest Model Results

Citycode	Adcode	Predicted Average Price (USD)
431	220104	8429.299359
431	220173	8334.973388
431	220103	6639.146611
431	220106	6771.753483

Table 5 GBR Model Results

Citycode	Adcode	Predicted Average Price (USD)
431	220104	8980.339255
431	220173	8566.413527
431	220103	6248.714788
431	220106	6948.823878
431	220183	3696.665550



picture 7 Future Housing Price Predictions in City1and City2

Results Analysis: The comparison of housing price predictions between city1 and Hohhot shows significant differences. In city1, housing prices are evenly distributed, with minimal gaps in economic development and public service levels among districts. The city's diversified industries contribute to stable housing prices. The Random Forest and Gradient Boosting Regression (GBR) models perform well in city1, reflecting the stable relationship between housing prices and various features. In Hohhot, housing prices show large regional disparities, with resource-driven industries causing high prices in certain areas and lower prices in remote regions. The Decision Tree model effectively captures these nonlinear relationships. For future

development, city1 should focus on balanced regional development and providing support for low-income families. Hohhot should aim to control high housing prices, promote development in low-priced areas, and ensure equity in public service distribution.

5.2 Test the model of question 2

5.2.1 Calculating Density Indices for the Two Cities

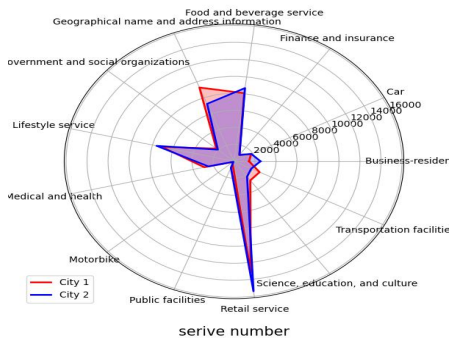
Table 6 The density indices of the two cities are as follows

server	City1	City2
Accommodation service	798.9943694	1013.089325
Business-residential	1435.372563	2523.326894
Car	1841.307732	1864.650652
Finance and insurance	902.3234561	914.0807105
Food and beverage service	8072.205387	8680.136041
Geographical name and address information	9285.952804	7228.189472
Government and social organizations	2253.676478	2065.405042
Lifestyle service	7092.946141	7595.47329
Medical and health	2925.080968	2517.331332
Motorbike	130.9099457	85.07762845
Public facilities	374.2111013	820.8897279
Retail service	13153.96229	15409.6803
Science, education, and culture	2729.742158	2194.805546
Transportation facilities	2752.644335	1857.519299

Based on the comparative assessment of density indices between the two cities, City 1 boasts a more developed tourism industry and vibrant commercial activities, with higher levels of economic development and better living services. City 2, on the other hand, indicates a broader urban coverage, greater convenience in transportation, and satisfactory living service quality, making it more suitable for long-term residence and raising children. The differences in other services are minimal. This suggests that both cities demonstrate balanced development, as well as similar social structures and populations.

5.2.2 Visualizing Density Indices with Radar Charts

To better understand the differences, we analyzed the density indices using radar charts, as shown in the figure below:



Picture 8 Density Index Radar Chart of city1 and city2

The radar chart provides a clear visualization of the focus areas for both cities. It

highlights that both cities place significant emphasis on the development of technology and cultural environments. However, there is room for improvement in economic growth and infrastructure development.

Using Python code, we calculated the coefficients of variation (Cv) for each service domain and converted them into weights. The results are shown in the table below:

Table 7 Coefficient of Variation and Weight Calculation

server	CV_value	Weight_value
Accommodation service	0.118148492	0.070031181
Business-residential	0.274826201	0.162900119
Car	0.006298754	0.003733515
Finance and insurance	0.006472818	0.00383669
Food and beverage service	0.036289295	0.021510069
Geographical name and address information	0.124606128	0.073858872
Government and social organizations	0.043590619	0.025837846
Lifestyle service	0.034212473	0.020279056
Medical and health	0.074920755	0.044408429
Motorbike	0.212198861	0.125778472
Public facilities	0.373758109	0.221540888
Retail service	0.078971651	0.046809552
Science, education, and culture	0.108626547	0.064387156
Transportation facilities	0.194163398	0.115088156

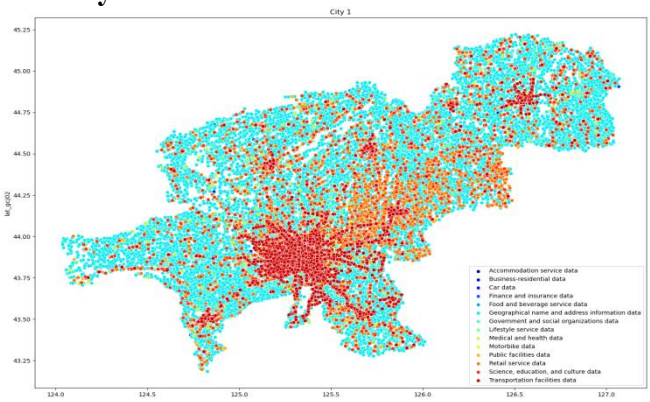
5.2.3 Comprehensive Scoring Using Weighted Density Indices

By assigning the calculated weights to each city's service density indices and applying the Multi-Criteria Evaluation (MCE) method with weighted summation, we obtained the following comprehensive scores:

	City 1	City 2
scores	4725.87	2801.21

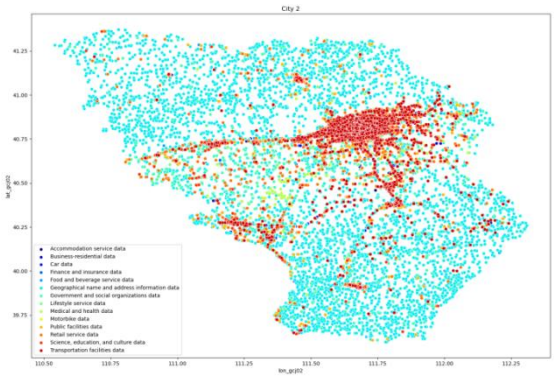
Based on these scores, we can conclude that City 1 has a significantly higher overall service level compared to City 2.

5.2.4 Visualization Analysis



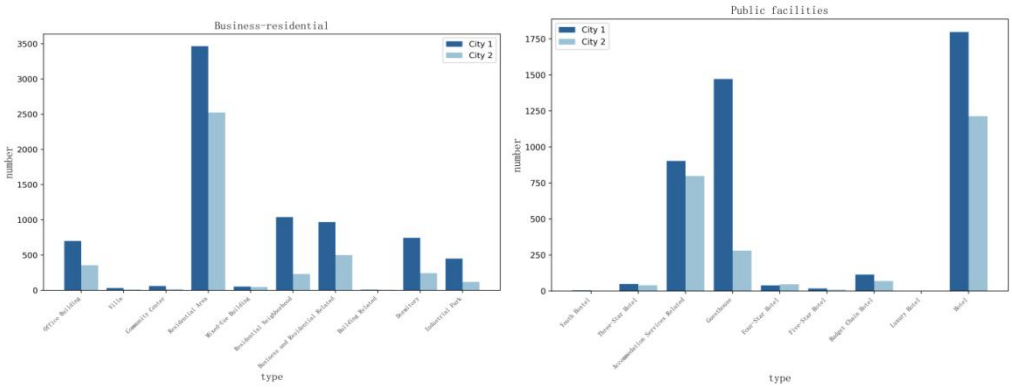
Picture 9 City 1 Facility Distribution Map

The distribution of service points in City 1 is dense and primarily concentrated in the city center, such as the Erdao District and Nanguan District, showcasing a highly urbanized characteristic with centralized services.



Picture 10 City 2 Facility Distribution Map

The distribution of service points in City 2 is relatively scattered, lacking a clearly defined city center. Public facilities and transportation infrastructure are distributed more evenly, but service coverage in peripheral areas shows significant deficiencies.



Picture 11 Bar Chart Comparison of Specific Service Data

We randomly selected two service comparison charts, and it is evident that the number of services in City 1 is greater than in City 2 across all categories.
This aligns with the comprehensive scores calculated through our model: City 1's score of 4725.87 significantly surpasses City 2's score of 2801.21, also confirming that City 1 has a higher overall service level than City 2.

5.2.5 Model Results

(1)Common Features

Both cities exhibit high density indices for retail services and food and beverage services, indicating that their economic development heavily relies on these sectors.

The density of medical services is also high in both cities, reflecting a strong emphasis on public health and healthcare infrastructure.

(2)Unique Features and Advantages/Disadvantages Analysis

City 1:

Advantages: City 1 leads in commerce and urban informatization, with relatively balanced distribution of transportation and public facilities, indicating more evenly distributed infrastructure investments.

Disadvantages: Service points are insufficiently distributed in suburban areas, resulting in a significant urban-rural divide.

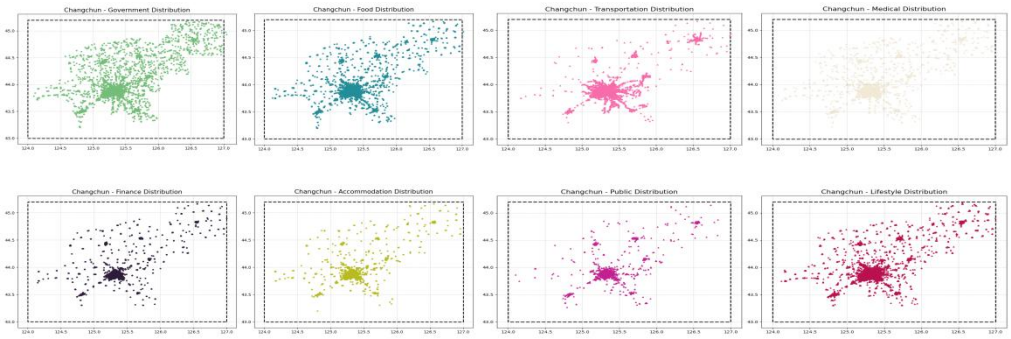
City 2:

Advantages: City 2 offers greater convenience in daily life for residents. The density of medical and health services is relatively high, showing strong attention to healthcare infrastructure.

Disadvantages: The distribution of public facilities and transportation infrastructure is uneven. Service coverage in peri-urban areas is sparse, and the layout of infrastructure requires optimization. While retail service density is high, it is overly concentrated in central urban areas, limiting accessibility for residents in peripheral regions.

5.3 Test the model of question 3

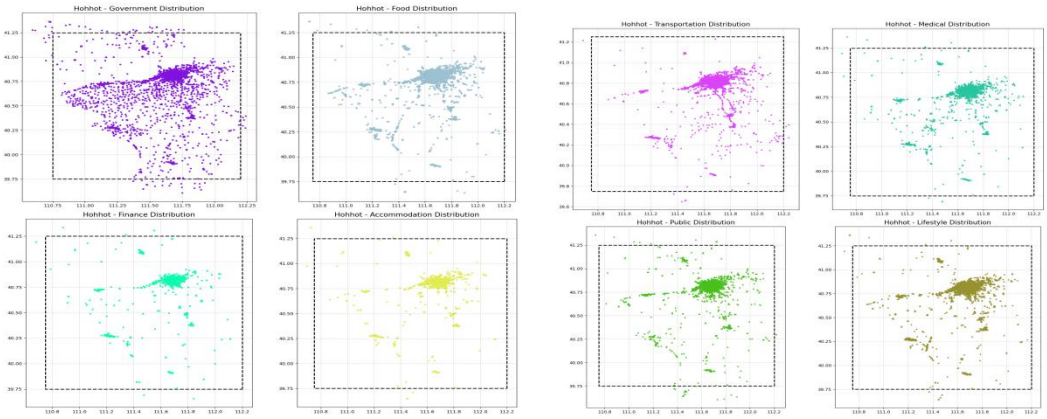
5.3.1 Calculate Scores of two cities



picture 12 City1 Resilience Capacity Distribution Map

Table 8 City1 Urban Resilience Scores in Various Aspects

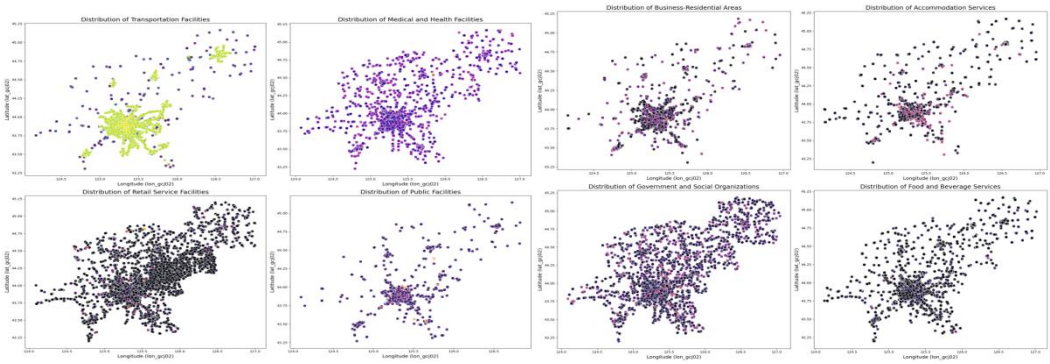
Server Facility	Weight	Reason	Score
Transportation	20%	Supports material transport and evacuation efficiency.	9
Medical Facilities	20%	Ensures health during and after emergencies.	9
Public Facilities	15%	Basic living security but less critical than others.	8
Lifestyle Services	10%	Meets basic living needs during emergencies.	8
Government & Orgs	10%	Coordinates resources and crisis management effectively.	8
Food & Beverage	10%	Supplies food during prolonged emergencies.	8
Finance & Insurance	5%	Aids post-disaster recovery but less direct impact.	5
Accommodation Services	10%	Provides shelters for affected individuals.	7



picture 13 City2 Resilience Capacity Distribution Map

Table 9 City2 Urban Resilience Scores in Various Aspects

Server Facility	Weight	Reason	Score
Transportation	20%	Centralized, sparse in outskirts, impacts rescue.	6
Medical Facilities	20%	Uneven, concentrated in city center, needs balance.	6
Public Facilities	15%	High density in center, lacks presence in outskirts.	5
Lifestyle Services	10%	Moderate density, better in center, insufficient outskirts.	6
Government & Orgs	10%	Centralized, limited coverage in some areas.	7
Food & Beverage	10%	Centralized, poor coverage in outskirts.	8
Finance & Insurance	5%	Limited, mostly in city center.	4
Accommodation Services	10%	Sparse, core area focus, needs better coverage.	5



picture 14 City1 Infrastructure Distribution Map

Table 10 City1 Sustainability Scores in Various Aspects

Server Facility	Weight	Reason	Score
Transportation	20%	Dense in center, sparse in outer areas.	6
Medical Facilities	18%	Even distribution but room for scattered improvements.	7
Retail Services	15%	Widely available, supports economic activity.	8

Public Facilities	10%	Low density, especially in outskirts.	5
Mixed-Use Areas	12%	Good development in core, underdeveloped in outskirts.	6
Accommodation Services	8%	Limited coverage in certain areas.	6
Government & Orgs	10%	Broad distribution but lacks coverage in outskirts.	7
Food & Beverage	7%	Evenly distributed but improvement needed in outskirts.	7
Total Score			6.55

picture 15 City2 Infrastructure Distribution Map

Table 11 City2 Sustainability Scores in Various Aspects

Server Facility	Weight	Reason	Score
Transportation	20%	Dense in center, sparse in outer areas.	6
Medical Facilities	18%	Even distribution but room for scattered improvements.	7
Retail Services	15%	Widely available, supports economic activity.	8
Public Facilities	10%	Low density, especially in outskirts.	5
Mixed-Use Areas	12%	Good development in core, underdeveloped in outskirts.	6
Accommodation Services	8%	Limited coverage in certain areas.	6
Government & Orgs	10%	Broad distribution but lacks coverage in outskirts.	7
Food & Beverage	7%	Evenly distributed but improvement needed in outskirts.	7
Total Score			5.35

5.3.2 Overview of the model

(1)Overview of city1

Existing Issues: Dense transportation facilities in central areas but sparse in outer areas, affecting the travel of some residents. Public facilities are limited in outer areas, resulting in unequal resource distribution. Mixed-use development is concentrated in

central areas, with underdeveloped outer areas leading to commuting pressure.
Development Direction: Expand outer transportation lines and optimize the network structure to improve public transit coverage. Increase parks and community centers in outer areas as public facilities. Promote mixed-use development and guide construction in outer areas.

Investment Strategy:
Short-term: Focus on optimizing the transportation network from the center to the outskirts, adding public transit routes, and improving roads. Prioritize constructing public facilities in densely populated outer areas.
Long-term: Plan comprehensive development zones in outer areas to promote mixed-use development. Simultaneously, develop green transportation such as rail and rapid transit to promote sustainable development.
(2)Overview of Hohhot

Existing Issues: Transportation is concentrated in the core areas, causing inconvenience and high commuting costs for outer areas. Medical resources are centralized, making it difficult for residents in remote areas to access healthcare. Some areas lack sufficient accommodation, affecting tourism and economic development.
Development Direction: Balance transportation layout, strengthen outer area development, and reduce urban-rural disparities. Build more medical facilities in suburban and remote areas to ensure healthcare access for residents. Improve accommodation and dining facilities around tourist attractions to enhance tourism appeal.

Investment Strategy
Short-term: Optimize transportation from the core to the outskirts by adding routes and maintaining roads. Establish small medical stations or clinics in remote areas as a priority.
Long-term: Build regional transportation hubs to improve efficiency and promote regional economic and population aggregation. Strengthen investment in tourist attraction infrastructure to boost tourism and increase urban revenue.
Comparison of Investment Priorities Between the Two Cities
Short-term: city1 prioritizes improving transportation and outer public facilities; Hohhot focuses on transportation improvements and adding medical facilities.
Long-term: city1 promotes mixed-use development in outer areas and green transportation construction; Hohhot focuses on building regional transportation hubs and improving tourism infrastructure.

6.Strengths and Weakness

6.1 Strengths

Problem 1: Housing Demand Prediction
Multiple Linear Regression: Simple and efficient, suitable for capturing linear

relationships in housing supply and demand, and easy to interpret the impact of features on demand fluctuations.
Gradient Boosting Regression (GBR): Provides high predictive accuracy, captures subtle trends in housing demand, and is robust to noise and outliers.
Problem 2: Urban Service System Analysis
Multi-Criteria Evaluation (MCE): Quantifies multiple service indicators (e.g., healthcare, transportation), provides a clear view of regional service balance, and is intuitive for policy optimization.
Problem 3: Urban Resilience Evaluation
Key Infrastructure Analysis: Identifies weak points in housing, transportation, and other critical infrastructure, optimizing emergency resource allocation and improving urban resilience.
Extreme Weather Evaluation: Provides predictive insights based on historical data, guiding resilience-building strategies for housing and infrastructure recovery.

6.2 Weaknesses

Problem 1: Housing Demand Prediction
Multiple Linear Regression: Cannot capture complex non-linear relationships and is sensitive to outliers in demand data.
Random Forest Regression: Complex and time-consuming, lacks interpretability of how features specifically influence housing demand fluctuations.
Gradient Boosting Regression (GBR): Computationally expensive, prone to overfitting, and difficult to interpret, making it less suitable for direct policy recommendations.
Problem 2: Urban Service System Analysis
Multi-Criteria Evaluation (MCE): Relies on subjective weight assignment, which may lead to biased optimization suggestions, and struggles to analyze complex interactions between service system indicators.
Problem 3: Urban Resilience Evaluation
Key Infrastructure Analysis: Depends on accurate and comprehensive data, overlooking non-infrastructure factors like population movement that affect emergency response.

7.Conclusion

7.1 Development Strategies for Question 4

The following is the future development plan based on the official statistical data of city1 (www.tjj.jl.gov.cn) and city2 (www.huhhot.gov.cn) on public service expenditures in recent years. These plans are made in combination with the testing

results of the models used in Questions 1, 2, and 3.

Table 12 Relevant Financial Expenditure Data for City1 and City2

Expenditure Category	City1 (\Billion Yuan)	City2 (\BillionYua)
Total	1074.97	581.35
Education	67.83	67.83
Health and Wellness	41.69	41.69
Housing Security	19.02	19.02
Transportation	32.15	32.15
Disaster Prevention and Emergency Management	2.84	2.84
Urban and Rural Community	120.77	120.77

7.1.1 Future Development Plan for City1

(1)Housing Demand Fluctuation Response

Investment Field: City1 should focus on optimizing housing supply in the city's peripheral areas to address the housing shortage in suburban regions.

Investment Amount: A short-term investment of 60 million yuan (approximately 30,000 square meters) will be allocated for rapid suburban housing construction to meet peak demand, and a long-term investment of 200 million yuan will be used to establish a smart housing management system to monitor housing demand in real time.

Expected Improvement: The current total housing stock is approximately 5.33 million square meters. Short-term investments will accelerate suburban housing construction, improving the supply-demand matching rate by 15%-20%. Long-term investment in the smart housing management system is expected to reduce price fluctuation ranges to within 10%.

(2)Service System Optimization

Investment Field: Enhance public service facilities in suburban and peripheral urban areas, improving the accessibility of transportation systems.

Investment Amount: A short-term investment of 1.5 billion yuan will be made to expand suburban public facilities, while a long-term investment of 3 billion yuan will be allocated to develop a smart transportation system.

Expected Improvement: The increase in public service facilities and smart transportation systems in suburban and peripheral areas will significantly enhance public service levels across City1, reducing regional development disparities.

(3)Urban Resilience Building

Investment Field: Strengthen the emergency response capacity of urban and peripheral areas while improving green infrastructure.

Investment Amount: A short-term investment of 20 million yuan will improve emergency response infrastructure, adding at least five comprehensive emergency shelters in key areas. A long-term investment of 400 million yuan will develop green infrastructure and an early warning system.

Expected Improvement: Enhanced emergency response and green infrastructure construction will make City1 more resilient, enabling quicker recovery from extreme weather and unexpected incidents.

7.1.2 Future Development Plan for City2

(1)Housing Demand Fluctuation Response:

Investment Field: City2 should focus on balancing housing supply between core and peripheral areas, addressing the housing shortages in outer regions.

Investment Amount: A short-term investment of 65 million yuan will be made for housing construction in peripheral areas, while a long-term investment of 180 million yuan will enhance housing management and price monitoring systems.

Expected Improvement: The current total housing stock in City2 is approximately 2.88 million square meters. Peripheral housing construction and the introduction of smart management systems will improve the supply-demand matching rate by 13%, ensuring City2 can effectively address future housing demand fluctuations. The price monitoring system is expected to reduce price fluctuation ranges to 8%-12%.

(2)Service System Optimization:

Investment Field: City2 should prioritize the layout of transportation and medical facilities, especially in suburban and peripheral areas.

Investment Amount: A short-term investment of 1.3 billion yuan will construct suburban medical and transportation infrastructure, while a long-term investment of 3 billion yuan will upgrade smart city facilities.

Expected Improvement: Accelerating the construction of suburban transportation and medical facilities, combined with the integration of smart city technologies, will increase medical service coverage from 75% to 85%, thereby improving public service satisfaction.

(3)Urban Resilience Building:

Investment Field: City2 should prioritize enhancing infrastructure resilience in peripheral areas and improving post-disaster recovery capacity.

Investment Amount: A short-term investment of 25 million yuan will establish a pilot emergency platform to improve emergency functionality in public facilities. A long-term investment of 400 million yuan will be allocated to developing green energy infrastructure, focusing on new wind and solar energy facilities.

Expected Improvement: Strengthened emergency response facilities and green infrastructure construction will increase shelter capacity by 20%, covering more remote areas. The proportion of renewable energy is expected to rise from the current 13% to 35%-40%.

In conclusion, this study analyzed housing markets, service systems, and urban resilience in two cities using data-driven methods. The Gradient Boosting model achieved high accuracy (R^2 = 0.9269) in predicting housing prices, while a Multi-Criteria Evaluation revealed significant service level disparities, with City 1 outperforming City 2. Resilience assessments highlighted City 1's strengths in infrastructure and City 2's weaknesses in peripheral facilities. Strategic short- and long-term investment plans were proposed to address these issues, promoting balanced and sustainable urban development.

References

[1] XIA Chenhong, ZHU Junyi, WANG Wei, GUO Xiaodong, MA Donghui. Demand for Systematic Risk Resilience Management and Planning Response Strategy in Megacities [J/OL]. Journal of Urban Sciences, 2024-11-13

[2] ZHANG Ke. Research on Resilience Evaluation of Urban Medical System Based on Entropy Weight Method-TOPSIS [D]. Beijing University of Civil Engineering and Architecture, 2022-6-10

Appendix

Details of the code used in this article can be found in”Python Code and Data” which includes:
(1)Question1Data Preprocessing
(2)Question1code
(3)Question2 code
(4)Question3 code
(5)House Price Prediction Results